This probability distribution is visualized in Figure 6.14. Verify that this is a valid probability distribution by proving that the probabilities sum to one. That is, prove that $Pr(X = 1) + Pr(X = 2) + \ldots + Pr(X = 10) = 1$.

2. Calculate the entropy of this prior distribution using the formula given at the end of Section 8.1.1.
3. Given the prior distribution defined between 1 and 10, what is the probability that the phone will break when dropped from the 2nd floor? What is this probability for the 5th floor? The 1st floor?
4. Assume that after observing the result of any trial, the posterior distribution for X is once again exponential and is defined between the lowest and highest possible floors.

   For example, if you observe that the phone doesn't break when dropped from the 5th floor, then we know that X is at least 5 and the posterior distribution of X is such that $Pr(X = 5) = 1 / 2$, $Pr(X = 6) = 1 / 4$, …, $Pr(X = 9) = 1 / 32$, $Pr(X = 10) = 1 / 32$. If, on the other hand, the phone breaks from the 5th floor, then we know that X is at most 4 and the posterior distribution is such that $Pr(X = 1) = 1 / 2$, $Pr(X = 2) = 1 / 4$, $Pr(X = 3) = 1 / 8$, $Pr(X = 10) = 1 / 8$.

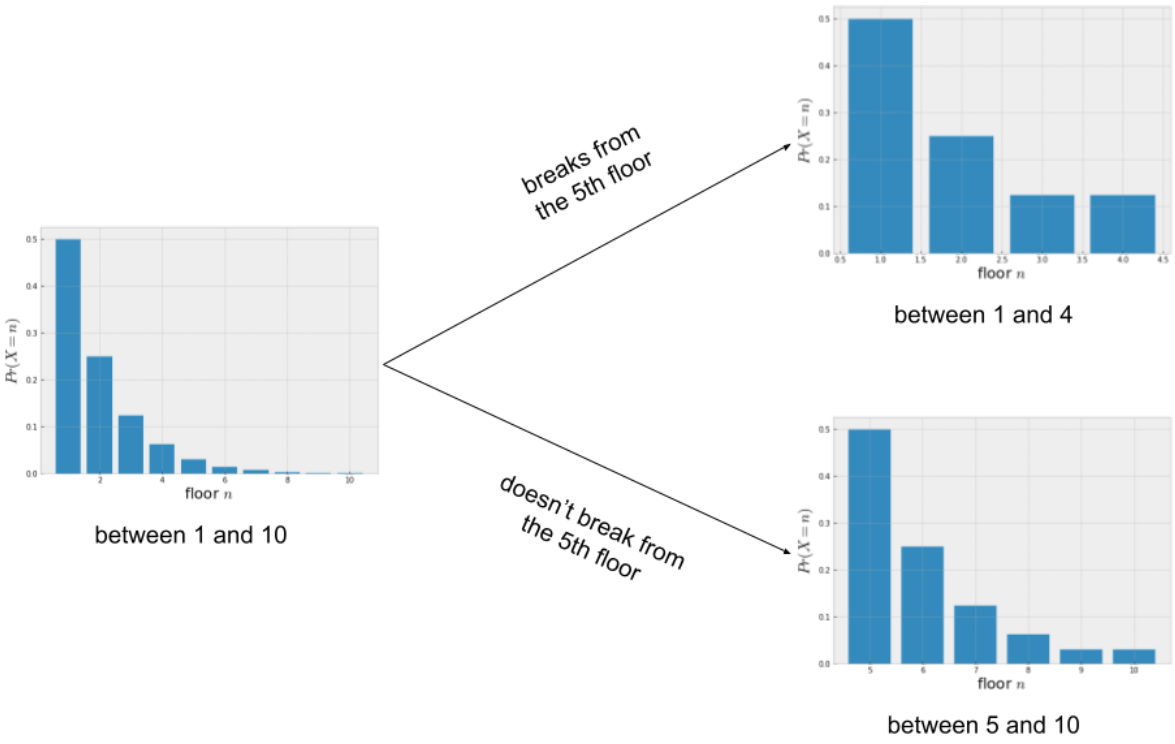


Figure 6.15 The posterior probability distributions of X in two scenarios when the phone is dropped from the 5th floor. Each posterior distribution is still exponential.

   Compute the entropy of this fictitious posterior distribution in the two cases.

5. Given the prior distribution, compute the expected posterior entropy after you conduct a trial on the 5th floor (that is, after you drop the phone from the 5th floor and observe whether it breaks).
6. Compute this expected posterior entropy for other floors. Which floor gives the highest reduction in entropy? Is this still the same result as binary search? If not, what has

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